Pythagorean Theorem in Real-World Problems

Teacher's Edition

The Period

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must Find a missing side in a real situation: the right angle located, the hypotenuse named, then the theorem; the answer in the story's unit, rounded once where it measures.

Must not 3-D (a box's space diagonal — the book's 6.4.5); radicands over 225; coordinates (6.05–06).

Target I can find a length in a real situation by finding the right triangle inside it.

MTR **MTR.7.1** — every board is a real object: a ladder, a ramp, a screen, a lawn, a kite · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.6.1** — boards 7 and 9: the answer is checked against the story

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Sketch the story and mark the square corner before any number.
2. A ladder, a diagonal, a ramp, a kite string: each is the hypotenuse.
3. How high, or how far along the ground, is a leg: subtract.

Slide by Slide

1. **Pythagorean Theorem in Real-World Problems.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's first move: every rectangle — a screen, a lawn, a box — has four right angles, so its diagonal is a hypotenuse. *1 retrieves 6.03 — a missing leg: subtract the squares (yesterday). 2 retrieves 6.01–02 — the converse, on the longest side (last week). 3 retrieves 5.04 — a real length from a plan's scale (last unit). 4 retrieves Grade 3–4 (MA.4.GR.1) — a rectangle's corners are right angles; today every diagonal cuts off a right triangle (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 7 ft 2. Yes — 81 + 1,600 = 1,681 = 41^2 3. 14 ft 4. 4
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the four triangles' names.** Sketch it. Mark the square corner. Name c.
6. **Notes — A garden 5 m by 12 m, and a path from corner to corner.** A rectangle's diagonal is always a hypotenuse.
7. **Notes.** How high is a leg. Subtract.
8. **Example 1 — A ramp's length.** Ask where the square corner is before any number: under the high end, where the rise meets the ground.
9. **Example 1 — Worked.** The rise and the run are the legs; the ramp is c.
10. **Your Turn — Same steps, your numbers. Boards up when done.** A screen's corner is square, so the diagonal is c: 49 + 100 = 149 and √149 ≈ 12.21, so 12.2 in — a board with 17 added the sides.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — How high a ladder reaches.** Ask which length is c before anyone squares: the ladder, 12 ft.
13. **Example 2 — Worked.** The ladder is c: h^2 = 144 − 25 = 119, √119 ≈ 10.91, so 10.9 ft — less than 12, as a leg must be.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 169 − 36 = 133 and √133 ≈ 11.53, so 11.5 ft — a board with 14.3 added the squares (√205), higher than the ladder is long.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A tablet screen is 6 inches wide and 8 inches tall. How long is its diagonal? → ***10 in*** *14 — added the sides [MA.8.GR.1.1: the theorem is about squares]*

18–19. Board 2. A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall. How high up the wall does it reach? → ***12 ft*** *17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]*

20–21. Board 3. A ramp rises 4 ft over a run of 11 ft. How long is the ramp, to the nearest tenth of a foot? → ***About 11.7 ft*** *15 — added the rise and the run [MA.8.GR.1.1]; '11, the same as the run' — the ramp is the hypotenuse, longer [Notes I]*

22–23. Board 4. A 20-ft ladder reaches 16 ft up a wall. How far from the wall is its foot? → ***A — 12 ft*** *(B) added the squares: about 25.6 ft, farther than the ladder is long*

[Notes III: for a leg, subtract]; (C) subtracted the lengths, 20 − 16 [MA.8.GR.1.1: the theorem is about squares]; (D) stopped at $a^2 = 144$ and forgot the square root [Math Nation names skipping the final root] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

24–25. Board 5. A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner. How much shorter is Ava's walk? → ***6 m shorter*** *15 — Ava's walk, not the difference [the question asks how much shorter]; 21 — Ben's walk*

26–27. Board 6. A kite flies on 30 m of string, straight above a spot 18 m from where the string is held. Which length is the hypotenuse? → ***C — the 30-m string*** *(A) took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (B) took the height as c — the height is the unknown leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) did not see the right angle under the kite [Florida B1G-M, GR.1.1: failing to visualize the triangle in a real-world problem] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A box is 7 inches wide and 12 inches long. Will a 14-in umbrella lie flat in it, corner to corner? → ***No — the diagonal is about 13.9 in.*** *'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]; 'Yes, 12 < 14' — compared with one side [Notes II: the diagonal is longest]*

30–31. Board 8. A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall. How high does it reach, to the nearest tenth of a foot? → ***About 7.1 ft*** *The 25 pounds is not needed. 12.2 — added the squares (√149): higher than the ladder is long [Notes III]*

32–33. Board 9. A window is 12 ft above the ground. Why can an 11-ft ladder never reach it, wherever its foot stands? → ***The height is a leg — always shorter than the 11-ft ladder.*** *'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a diagonal and a ladder's height, both whole; 3 a ramp to the tenth; 4 the ladder's foot, voted. The pattern breaks at 5 — two walks and their difference — and 6, which asks only which length is c; 7 is will-it-fit by a tenth; 8 has the extra number; 9 is the reason in writing. The independent set: a diagonal, a ladder's foot, a ramp, along two sides against across, will it fit, a kite's height.

Misconceptions to Watch

Board 1. 14 — added the sides [MA.8.GR.1.1: the theorem is about squares]

Board 2. 17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]

Board 3. 15 — added the rise and the run [MA.8.GR.1.1]

Board 4. added the squares: about 25.6 ft, farther than the ladder is long [Notes III: for a leg, subtract]

Board 5. 15 — Ava's walk, not the difference [the question asks how much shorter]

Board 6. took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]

Board 7. 'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]

Board 8. 12.2 — added the squares (√149): higher than the ladder is long [Notes III]

Board 9. 'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]

Materials Whiteboards and markers. A meter stick or a ladder picture for Notes I if the room needs it. Calculators allowed, printed nowhere.